GATE

Computer Science And Information Technology

C-Programming

Arrays & Pointers

DPP: 1

Q1 Which of the following declarations are INVALID?
(A) int b[][4];
(B) int b[];
(C) int b[2][][2]={1,2,3,4};
(D) int b[][2][2]={1,2,3,4};

Q2 Consider the following two statements:
P: int a[3]={1, 2, 3};
printf("%d", *a++);
Q: int a[3]={1, 2, 3};
int *p=a;
printf("%d", *p++);
Which of the following statements is/are CORRECT?
(A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q

Q3 Consider the following program:

```
#include<stdio.h>
int main(void)
{
    int a[5]={5, 10, 15};
    printf("%d", 1[a]);
    return 0;
}
```

The output is-
(A) 5
(B) 10
(C) Garbage value
(D) compilation error

Q4 Consider the following program:

```
#include<stdio.h>
int main(void)
{
    int 5[a]={5, 10, 15};
    printf("%d", 1[a]);
    return 0;
}
```

The output is-
(A) 5
(B) 10
(C) Garbage value
(D) Compilation error

Q5 Consider the following program:

```
#include<stdio.h>
int main(void)
{
    int a[5]={5, 10, 15, 20, 25};
    printf("%u", a);
    printf("%u", *(a+3));
    printf("%u", a+2);
    printf("%u", *(a+2)+6);
    printf("%u",*(a+*(a+1)-6));
    return 0;
}
```

Assuming the base address of the array to be 1000 and integer size as two bytes the output is-
(A) 1000 20 1004 21 25
(B) 5 20 15 21 25
(C) 1000 20 1002 21 24
(D) Compilation error

Q6 Consider the following program:

```
#include<stdio.h>
int main(void)
{
    int a[5]={5, 10, 15, 20, 25};
    printf("%u\t", *(1+a));
    printf("%u\t", &a+1);
    return 0;
```

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}

Assuming the base address of the array to be 1000 and integer size as four bytes the output is-

(A) 1004 1020
(B) 10 1016
(C) 10 1020
(D) 1004 1016

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Answer Key

Q1	(A, B, C)	Q4	(D)
Q2	(B)	Q5	(A)
Q3	(B)	Q6	(C)

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Hints & Solutions

Q1 Text Solution:

(a) int b[][4]: Invalid as elements are not specified.

(b) int b[]; Invalid as size is not specified.

(c) int b[2][][2]={1,2,3,4}; Invalid. If the elements are specified, only first dimension can be omitted.

(d) int b[][2][2]={1,2,3,4}; Valid. If the elements are specified, only first dimension can be omitted.

Q2 Text Solution:

int a[3]={1, 2, 3};

Array name without subscript denotes the base address of the array. So, a++ is not allowed.

Hence, P is incorrect.

Q is correct.

Q3 Text Solution:

The printf() statement can be interpreted as- printf("%d", 1[a]) is equivalent to printf("%d", *(1+a));

1000	1002	1004
5	10	15

So, *(1+a) is equivalent to *(1+1000). Here, 1 signifies the increment by 1*2 bytes= 2 bytes.

So, *(1002) is 10.

Q4 Text Solution:

int 5[a]={5, 10, 15}; // It is an invalid declaration.

So, compilation error will happen.

Q5 Text Solution:

1000	1002	1004	1006	1008
5	10	15	20	25

printf("%u", a);//1000

printf("%u", *(a+3));//*(1000+2*3) i.e *1006 i.e 20

printf("%u", a+2);//1000+2*2=1004

printf("%u", *(a+2)+6);//*(1000+2*2)+6 i.e *1004+6 i.e 15+6 i.e 21

printf("%u",*(a+*(a+1)-6));

//*(a+*(1000+2*1)-6) = *(a+4) = *(1000+2*4) = *1008 i.e 25

Output: 1000 20 1004 21 25

Q6 Text Solution:

1000	1004	1008	1012	1016
5	10	15	20	25

printf("%u\t", *(1+a));//*(1*4+1000)=*1004=10

&a+1 signifies the next 1D array. So, size incrementing by 1 means increase by 4*5 bytes.

printf("%u\t", &a+1);//1000+20=1020 is printed.

Output: 10 1020

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